Chaski · measured results

Hands-free Quechua Chanka – Spanish medical interpreter · Cal Hacks 13.0

Sunday, October 25, 2026

Every number below comes from a results file in the repo (`github.com/Freddyx14/chaski`), measured on the live service.

What	Result	Source
Seeded errors caught before speaking (952-case audited bank, 224-case run)	**183/184 (99.5 %), 100 % of dose, day, negation, medicine, unit and «with water» errors**	`evals/ataques/`
Correct sentences blocked by mistake	7/40 (17.5 %), about half genuinely weak family translations; a block asks the speaker to repeat	same
Naive "is this right?" judge (Sonnet), same sample	catches errors but blocks 27.5 % of correct sentences and can't say which fact failed	`evals/ataques/`
Whole consultations in hands-free mode (22 turns, nobody tells Chaski who speaks; real Quechua from a speaker the recognizer never heard)	who spoke **21/22** · right person and language **18/18** · **0 numbers lost** · 7.8 s per turn · 6–9 turn consultation in 49–70 s	`evals/conversacion/`
Real Quechua audio, held-out speaker, 30 turns	24 said, 6 asked to repeat, 0 blocked, numbers intact 3/3	`evals/vivo/`
Interpreter picks the right action (48 real scenarios)	Claude Opus 5: 98 %, 0 invented clinical data, 2.3 s	`evals/interprete/`
Translator choice (62 unseen family phrases, ~5 references each)	Claude Sonnet 5 damaged 0 of 92 critical terms; NLLB-1.3B 3, Gemini 3 Flash 4	`evals/traduccion/`
Community nurse's reviewed phrases as examples	chrF++ Spanish→Quechua 52.6 → 56.0, Quechua→Spanish 49.1 → 56.0 (leakage-guarded)	`evals/moat/`
Fine-tuned Quechua speech recognition (held-out speaker)	word error rate 53.9 % → 40.9 %	commit `7b57a9b`
Live latency (`/turno/vivo`)	who spoke 0.5–1 s · text on screen 3–5 s · voice ready ~8.5 s	`evals/HISTORIA.md`
Cost	~$0.02 per translated turn (list prices)	`evals/latencia/`

What	Result	Source
Fresh-context safety audit	1 critical, 3 high found and fixed; seeded-error numbers unchanged after the fixes	`evals/auditoria/`

Design rule: the model proposes, code decides. Nothing doubtful reaches the patient: if a check fails, Chaski asks the person who spoke to repeat, naming the fact it wasn't sure of.